Classifier Selection Method Based on Multiple Diversity Measures

KeFei Cheng, ZhiWen Song, YaNan Yue, FengChi Shan, XiaoYong Guo

College of Computer Science & Technology, Chongqing University of Posts and Telecommunications

Chongqing 400065, China

chengkf@cqupt.edu.cn, songzhiwenemail@qq.com

Abstract—**Multiple Classifier Systems can make up for some defects of a single classifier. It has been widely used in machine learning, pattern recognition, and other fields. However, it is easy to generate some redundant classifiers with small difference, when the number of classifiers increases. In order to select classifiers with great diversity, a classifier selection method based on multiple diversity measures is proposed in this paper. Firstly, the fusion matrix is constructed by using five pairwise diversity measures. Then, the graph obtained by the fusion matrix is colored by the ant colony algorithm, and the candidate ensembles are generated. Finally, we introduce the fuzzy information theory and combine with five non-pairwise diversity measures to select a group of classifiers. The experimental results show that the proposed method is feasible and can significantly improve the accuracy of the ensemble.**

Keywords-multiple classifier systems; diversity measure; ant colony algorithm; graph coloring; fuzzy information entropy

I. INTRODUCTION

Classifier ensemble is a method of combining multiple classifiers to improve classification accuracy and generalization ability. The research fields involved are not limited to machine learning and pattern recognition, but also extend to neural networks, genetic algorithms, and other multidisciplinary fields. However, the following problems will inevitably occur, when the scale of ensemble increases: 1) It takes lots of time to train multiple classifiers for ensemble; 2) The ensemble of multiple classifiers will occupy a huge storage space [1]; 3) As the number of classifiers increases, it is easy to generate redundant classifiers with little difference [2]. In order to solve these problems, Zhou et al. [3] proposed the concept of "selective ensemble" and studied it. The research result shows that better performance can be obtained by selecting classifiers with a certain degree of diversity and accuracy from multiple classifiers.

The research [4] shows that, the two important factors in selecting classifiers are independence and diversity. If the output of one classifier has no effect on another classifier, then the two classifiers are independent; if they output different results for the same training data, then the two classifiers are diverse. Some approaches [5] have been researched for achieving the maximum diversity between classifiers, the most popular are: subsampling the training data, partitioning the features of the training data, using different types of classifiers. For example, Bagging is a popular ensemble algorithm, which generates diverse classifiers through Bootstrap sampling disturbance training set. Random subspace method (RSM) is another popular ensemble algorithm that generates diverse classifiers by using some random features rather than all features. With the increase in the number of classifiers, these two methods will produce partially redundant classifiers. In order to solve this problem, some scholars began to study the selection of classifiers based on diversity measure. Kuncheva et al. [6] summarized 10 kinds of diversity measures and designed experiments to test the relationship between ensemble accuracy and diversity measure. Tang et al. [7] presented a theoretical analysis of six diversity measures and showed underlying relationships between them. Li et al. [8] proposed the Diversity Regularized Ensemble Pruning (DREP) method, which achieves significantly better generalization performance. Cavalcanti et al. [9] introduced an ensemble pruning method that combines multiple diversity measures by using a genetic algorithm and a graph coloring algorithm.

At present, the selection of classifiers based on diversity measure mainly has two aspects: take a single diversity measure as the standard or take a kind of diversity measures as the standard. Both of them might not be accurate enough to capture all the relevant diversity in the ensemble [10]. Therefore, a new classifier selection method is proposed in this paper, which considers both pairwise diversity measures and non-pairwise diversity measures. Firstly, five pairwise diversity matrices are calculated by five common pairwise diversity measures. The five matrices are merged into one matrix by voting. Then use the ant colony algorithm to color the graph obtained by the fusion matrix. Thus the classifiers are grouped. Finally, the fuzzy information theory [11] is introduced, and five non-pairwise diversity measures are combined to select an ensemble.

The rest of this article is organized as follows. In section 2, several commonly used diversity measures are briefly reviewed. In section 3, the proposed method is described in detail. Experimental results are given in section 4 to verify the

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effectiveness of the method. In section 5, the conclusion is given.

II. Common Diversity Measures

Diversity measure is used to measure the diversity of individual classifiers in ensemble. Pairwise diversity measure and non-pairwise diversity measure are two commonly used diversity measures. The following is a detailed description of the two diversity measures.

A. Pairwise Diversity Measure

Pairwise measures are calculated between two member classifiers in the classifier system. The diversity measure of the team is the averaged measures over all pairwise.

Suppose $D = \{d_1, d_2, \dots, d_L\}$represents L classifiers. The diversity measures are calculated using a contingency table that summarizes the behavior of two classifiers d_i and d_j across a dataset. Table 1 shows an example of a contingency table. The values on the table have the following meaning: x_1 is the number of examples in the dataset correctly classified by both d_i and d_j; x_2 is the number of examples correctly classified by d_i and incorrectly classified by d_j; x_3 is the number of examples incorrectly classified by d_i and correctly classified by d_j; x_4 is the number of examples incorrectly classified by both classifiers. Some common pairwise diversity measures are given below.

TABLE I. Contingency Table for Two Classifiers d_i and d_j

	$d_i = +$	$d_j = -$
$d_i = +$	x_1	x_2
$d_j = -$	x_3	x_4

1) The correlation coefficient ρ

$$\rho_{i,j} = \frac{x_1 x_4 - x_2 x_3}{[(x_1 + x_2)(x_3 + x_4)(x_1 + x_3)(x_2 + x_4)]^{\frac{1}{2}}} \quad (1)$$

2) The Q statistics

$$Q_{i,j} = \frac{x_1 x_4 - x_2 x_3}{x_1 x_4 + x_2 x_3} \quad (2)$$

3) The disagreement measure

$$D_{i,j} = \frac{x_2 + x_3}{x_1 + x_2 + x_3 + x_4} \quad (3)$$

4) The double-fault measure

$$DF_{i,j} = \frac{x_4}{x_1 + x_2 + x_3 + x_4} \quad (4)$$

5) The Kappa-statistic

$$K_{i,j} = \frac{\theta_1 - \theta_2}{1 - \theta_2} \quad (5)$$

Where

$$\theta_1 = \frac{x_1 + x_4}{x_1 + x_2 + x_3 + x_4} \quad (6)$$

$$\theta_2 = \frac{(x_1 + x_3)(x_1 + x_2) + (x_2 + x_4)(x_3 + x_4)}{(x_1 + x_2 + x_3 + x_4)^2} \quad (7)$$

B. Non-pairwise Diversity Measure

Non-pairwise measures are calculated directly over all member classifiers. The diversity of the team is the diversity of entire classifiers ensemble.

1) The entropy measure E

$$E = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{(L - [L/2])} min\{l(m_i), L - l(m_i)\} \quad (8)$$

Where, L is the number of classifiers. N is the number of training samples. $l(m_i)$ represents the number of classifiers that correctly classify the sample m_i.

2) Kohavi-Wolpert variance

$$KW = \frac{1}{NL^2} \sum_{i=1}^{N} l(m_i) * (L - l(m_i)) \quad (9)$$

3) The measure of "difficulty" θ

$$\theta = var(Z) \quad (10)$$

Where, Z represents the proportion of all classifiers that correctly classify a random sample m.

4) Generalized diversity

$$GD = 1 - \frac{P(2)}{P(1)} \quad (11)$$

Where, $P(1)$ represents the probability of classification errors for any one classifier and $P(2)$ represents the probability of classification errors for any two classifiers.

5) Measurement of interrater agreement k

$$K = 1 - \frac{\frac{1}{L}\sum_{i=1}^{N} l(m_i) * (L - l(m_i))}{N(L-1)\bar{P}(1-\bar{P})} \quad (12)$$

Where, $\bar{P}$ is the average individual classification accuracy in the ensemble.

III. Proposed Method

This section introduces the proposed method in this paper, which consists of three parts. The first part is the fusion of five pairwise diversity measures. The second part is to color the graph according to the ant colony algorithm. The third part is to combine five non-pairwise diversity measures and take fuzzy information entropy as the evaluation index. Fig. 1 depicts all the parts of the proposed method.

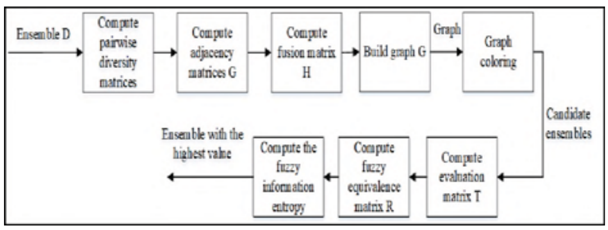

Figure 1. Outline of the proposed method

A. *Fusion of Pairwise Diversity Measures*

The initial pool of classifiers D of size L is created using Bagging. The five pairwise diversity measures reviewed in Section 2 were considered in this part: Correlation coefficient(P_1), Q-Statistics (P_2), Disagreement measure(P_3), Double-fault measure(P_4), and Kappa-statistic(P_5). These five pairwise diversity measures are used to calculate the diversity of each pair of classifiers, then five $L*L$ matrices are constructed. The matrix P is shown in (13).

$$P=\begin{bmatrix} d_{11}d_{12} & \dots & d_{1L} \\ d_{21}\, d_{22} & \dots & d_{2L} \\ & \vdots & \\ d_{L1}d_{L2} & \dots & d_{LL} \end{bmatrix} \tag{13}$$

Then we construct a fusion matrix based on the above five matrices, the steps are as follows:

1) *Calculate the average value of the diversity between classifier d_i and other classifiers*

$$A_i=\sum_{i=1}^{L} d_{ij}/L \tag{14}$$

2) *Calculate the average diversity of all classifiers.*

$$AV=\sum_{i=1}^{L} A_i/L \tag{15}$$

3) *Compare the diversity between classifiers d_i and d_j with AV, and construct five $L*L$ adjacency matrices G_{ij}.*

$$G_{ij}=\begin{cases}1 & d_{ij}\geq AV\\ 0 & d_{ij}<AV\end{cases} \tag{16}$$

4) *Five adjacent matrices are transformed into a fusion matrix H by voting.*

Equation (17) and (18) show examples of P and G_{ij} matrices, respectively. These matrices are computed using a pool of five classifiers. In this example, the threshold $AV=0.4$. Applying the matrix P to (16), we obtain the adjacency matrices G_{ij} showed in (18).

$$P=\begin{bmatrix} 0 & 0.4 & 0.8 & 0.3 & 0.5 \\ 0.4 & 0 & 0.7 & 0.5 & 0.4 \\ 0.8 & 0.7 & 0 & 0.3 & 0.7 \\ 0.3 & 0.5 & 0.3 & 0 & 0.4 \\ 0.5 & 0.4 & 0.7 & 0.4 & 0 \end{bmatrix} \tag{17}$$

$$G_{ij}=\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \tag{18}$$

Through five pairwise diversity measures, we can construct five adjacent matrices G_{ij} as shown above. Equation (19) is an example of a fusion matrix H obtained by voting. Building a graph G (Fig. 2) from the fusion matrix H. d_i and d_j are adjacent vertices, which have a low combined diversity and should not be in the same ensemble.

$$H=\begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{19}$$

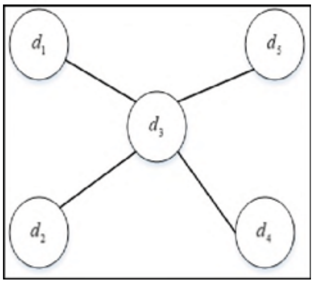

Figure 2. The graph G created from the fusion matrix H

B. *Graph Coloring Using Ant Colony algorithm*

Graph Coloring Problem (GCP) is a classic problem in graph theory. Given an undirected graph $G=(V,E)$, the problem is to find a coloring of the vertices with minimum number of colors and no pair of adjacent vertices has the same color. The minimal number of colors k for which a k-colouring exists is called the chromatic number of G.

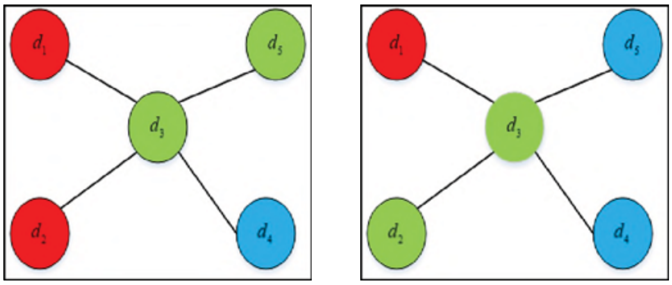

Figure 3. Two possible coloring schemes for graph G

Fig. 3 shows two schemes for coloring graph G, the candidate ensembles would be $\{d_1,d_2\},\{d_4\},\{d_3,d_5\}$ and $\{d_1\},\{d_2,d_3\},\{d_4,d_5\}$ respectively, and the chromatic number of G is 3. Because ant colony algorithm has the characteristics of good positive feedback, robustness and group [12], we use it to solve graph coloring problem in this paper.

Given an undirected graph $G=(V,E)$ and the vertices sets $V=\{v_1,v_2,\dots,v_L\}$. Suppose L vertices represent L classifiers, and L vertices are colored with m ants. ρ is the persistence of the trail, and α and β are two parameters, which determine the relative influence of pheromone trail and heuristic information. NC represents the number of iterations. $C=\{c_1,c_2,\dots,c_p\}$ is a color set. We give below the framework and the components of our approach MMAS for the GCP.

1) *Update of pheromone trails:*

In order to prevent the algorithm falling into the local optimal solution when it iterates to a certain number of times, we use an improved ant colony algorithm [13]. The algorithm only updates the pheromone after each cycle completed and the pheromone trail update rule is given by:

$$\tau_{ij}(t+1) = (1-\rho) * \tau_{ij}(t) + \Delta\tau_{ij}^{best} \tag{20}$$

Where

$$\Delta\tau_{ij}^{best} = 1/Numc^{best} \tag{21}$$

τ_{ij} is the value of pheromone when v_i is colored by c_j at the iteration "$t+1$", $Numc^{best}$ represents the current minimum number of colors during the iteration. The pheromone values are limited in the update procedure as follows:

$$\tau_{ij} = \begin{cases} \tau_{max} & \tau_{ij} \geq \tau_{max} \\ \tau_{min} & \tau_{ij} \leq \tau_{min} \end{cases} \tag{22}$$

Where

$$\tau_{max} = \frac{1}{1-\rho} * \frac{1}{Numc^{best}} \tag{23}$$

$$\tau_{min} = \begin{cases} \tau_{max}/20 & NC \leq 2 \\ \dfrac{\tau_{max}\left(1-\sqrt[NC]{P_{best}}\right)}{(avg-1)\sqrt[NC]{P_{best}}} & NC > 2 \end{cases} \tag{24}$$

Where, $avg = NC/2$, $P_{best} = 0.5$. In the beginning, the initial pheromone level is set to $\tau_{\max}$ instead of a constant.

2) The State Transition Rule

Move probability p_{ij}^k is defined as the probability that ant k colors v_i with c_j. It is computed using:

$$p_{ij}^k = \begin{cases} \tau_{ij}^{\alpha}\eta_{ij}^{\beta} / \sum \tau_{ij}^{\alpha}\eta_{ij}^{\beta} & if\ c_j \in allowed_i^k \\ 0 & else \end{cases} \tag{25}$$

Where, η_{ij} is the heuristic information, $allowed_i^k$ represents a color set that were already assigned to the adjacent vertices of v_i.

C. *Evaluation method of classifiers based on fuzzy information entropy*

After coloring the graph according to the ant colony algorithm, we get several candidate ensembles. In order to select one ensemble, we combine five non-pairwise diversity measures, and establish evaluation system based on fuzzy information entropy.

The common non-pairwise diversity measures are: the entropy measure (M_1), Kohavi-Wolpert variance (M_2), the measure of "$difficulty$"(M_3), Generalized diversity (M_4), Measurement of interrater agreement (M_5). These five non-pairwise diversity measures are used to calculate the diversity of each candidate ensemble, then we can construct an evaluation matrix T.

$$T = \begin{bmatrix} f_{M_{11}} f_{M_{12}} & \cdots & f_{M_{1n}} \\ f_{M_{21}} f_{M_{22}} & \cdots & f_{M_{2n}} \\ & \vdots & \\ f_{M_{51}} f_{M_{52}} & \cdots & f_{M_{5n}} \end{bmatrix} \tag{26}$$

Because the values calculated by these five methods are different in magnitude, we should normalize the values. The method is shown in (27).

$$f_{M_{ij}}^* = \frac{f_{M_{ij}} - m_j}{M_j - m_j}, i = 1,2,3,4,5 \quad j = 1,2,\ldots n \tag{27}$$

Where, $M_j = \max\left\{f_{M_{1j}}, f_{M_{2j}}, f_{M_{3j}}, f_{M_{4j}}, f_{M_{5j}}\right\}$, $m_j = \min\left\{f_{M_{1j}}, f_{M_{2j}}, f_{M_{3j}}, f_{M_{4j}}, f_{M_{5j}}\right\}, j = 1,2,\ldots,n$.

After values normalized, we can get the fuzzy similarity matrix T_r by (28).

$$r_{M_{ij}} = \frac{\sum_{k=1}^{n} f_{M_{ik}}^* \cdot f_{M_{jk}}^*}{\sqrt{\left(\sum_{k=1}^{n} f_{M_{ik}}^{*2}\right)\left(\sum_{k=1}^{n} f_{M_{ik}}^{*2}\right)}} \tag{28}$$

Definition 3.3.1: Suppose there is a fuzzy matrix R, if R satisfies:

- Reflexivity: $I \in R$;
- Symmetry: $R^T \in R$;
- Transitivity: $R^{\circ}R \in R$;

Then, R is called the fuzzy equivalence matrix. The similarity matrix T_r obtained above generally does not satisfy transitivity, so it is not a fuzzy equivalent matrix. We use square method to transform the similarity matrix into the fuzzy equivalent matrix.

Definition 3.3.2: Suppose A is a finite set of samples, F is a set of attributes of the sample, R is the fuzzy equivalence matrix of set A in attribute space F. As shown in (29).

$$R = \begin{bmatrix} f_{11} f_{12} & \cdots & f_{1m} \\ f_{21} f_{22} & \cdots & f_{2m} \\ & \vdots & \\ f_{51} f_{52} & \cdots & f_{5m} \end{bmatrix} \tag{29}$$

Then the fuzzy information entropy of the attribute set F is defined as:

$$F(R) = -\frac{1}{m}\sum_{i=1}^{m} log \frac{\|[a_i]_F\|}{m} \tag{30}$$

Where, $\|[a_i]_F\| = \sum_{j=1}^{m} f_{ij}$.

We use (30) to calculate the fuzzy information entropy of each candidate ensemble, and select one with the largest value as output ensemble.

IV. EXPERIMENTS

A. *Experimental Design*

In this experiment, 10 datasets are selected to verify the effectiveness of the method. The datasets are all from the public UCI datasets, as shown in table 2. We use the unstable decision tree as the base classifier. Majority vote was the fusion rule employed. In order to verify the effectiveness of the method in this paper, comparisons are made in the following ways:

1) *Comparison of the popular ensemble methods:*

Compare with the Bagging and Adaboost algorithms. The pool of classifiers was created using Bagging and the number of classifiers assumes three different values $L = \{50, 100, 150\}$.

2) *Comparison of selective ensemble methods:*

Compare with the DREP [8] and DivP [9] algorithms. We select the number $L = 100$ that produced the best results in the previous experiments.

3) *Comparison of different pairwise diversity measures:*

Compare the five popular pairwise diversity measures(P_1, P_2, P_3, P_4, P_5)with the pairwise fusion methods of this paper. Five pairwise diversity measures are used to replace our method respectively. Similarly, we select the number $L = 100$.

4) *Comparison of different non-pairwise diversity measures:*

Compare the five popular non-pairwise diversity measures(M_1, M_2, M_3, M_4, M_5) with the evaluation method of this paper. Five non-pairwise diversity measures are used to replace our method respectively. Similarly, we select the number $L = 100$.

A random method is adopted to divide 10% of the original data set into test set, 90% of the original data set into train set. The experiment is repeated for 10 times, and the average value is taken as the final evaluation result.

TABLE II. THE EXPERIMENTAL DATA SET

Abbreviation	Datasets	Sample	Attribute	Category
Wine	Wine	178	3	13
Aus	Australian	690	15	2
Veh	Vehicle	846	19	4
Iono	Inonosphere	351	35	2
Wf	Waveform	5000	21	3
Sonar	Sonar	208	61	2
Rwine	Wine_quality	1599	12	6
Gc	G_credit	1000	25	2
Sb	Spambase	4601	57	2
Eco	Ecoli	336	7	8

B. Experimental Results

As shown in table 3 at the last of paper, the classification performance of our method is better than Bagging and Adaboost algorithm on most datasets when the number of classifiers is 50, 100 and 150.

Compared with Bagging and Adaboost, the average accuracy of the proposed method is improved by 6.59% and 8.37% respectively when the number of classifiers is 50. The average accuracy of the proposed method is improved by 6.76% and 8.51% respectively when the number of classifiers is 100. The average accuracy of the proposed method is improved by 5.21% and 6.44% respectively when the number of classifiers is 150. The proposed method has the highest average classification accuracy and the best improvement effect when number is 100.

As shown in table 4 at the last of paper, the classification performance of the proposed method is better than DREP and DivP algorithm on most datasets when the number of classifiers is 100. These results show that the proposed method is feasible and effective.

Table 5 shows the results of our method with and without the fusion of the pairwise diversity measures. The combination of five pairwise diversity measures obtained the best accuracy rates in all except three datasets. These results show that the fusion of pairwise diversity measures is feasible.

From table 6, we can see that, with the exception of Iono, Wf datasets, the method in this paper is better than the single non-pairwise diversity measure, which shows the feasibility of the evaluation system based on fuzzy information entropy.

V. CONCLUSIONS

This paper proposes a classifier selection method based on multiple diversity measures. And the method combines pairwise diversity measures and non-pairwise diversity measures. Experimental results on 10 UCI datasets show the effectiveness and feasibility of the proposed method under the same experimental conditions.

It should be pointed out that the graph coloring algorithm used in this paper is the max–min ant colony algorithm. Subsequent attempts can be made to use other improved ant colony algorithms. In addition, the future work will also be studied on the parallelization of the ensemble system.

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TABLE III. COMPARE WITH THE BAGGING AND ADABOOST ALGORITHMS FOR DIFFERENT NUMBER OF CLASSIFIERS

	L=50			L=100			L=150		
Datasets	**My**	**Bagging**	**Adaboost**	**My**	**Bagging**	**Adaboost**	**My**	**Bagging**	**Adaboost**
Wine	**0.9722**	0.9074	0.8021	**0.9722**	0.9344	0.8921	**0.9722**	0.9174	0.8321
Aus	**0.8696**	0.6667	0.7709	**0.8841**	0.6815	0.7485	**0.8188**	0.6715	0.7046
Veh	**0.7588**	0.7087	0.6929	**0.7824**	0.7087	0.7313	**0.7347**	0.7087	0.6813
Iono	0.8732	**0.9239**	0.9106	0.8310	0.9140	**0.9287**	0.8197	**0.9340**	0.9062
Wf	**0.8600**	0.5613	0.7640	**0.8630**	0.5613	0.6014	**0.8290**	0.5607	0.7428
Sonar	**0.8095**	0.6825	0.7277	**0.8124**	0.6890	0.7495	**0.7869**	0.6667	0.7415
Rwine	**0.5625**	0.5624	0.5429	**0.6125**	0.5604	0.5829	**0.5878**	0.5663	0.5578
Gc	**0.8000**	0.7467	0.7614	**0.7800**	0.7567	0.7600	0.7450	**0.7633**	0.7600
Sb	**0.9153**	0.7777	0.8034	**0.9349**	0.7971	0.8250	**0.9262**	0.7835	0.8484
Eco	0.7647	**0.8118**	0.7508	0.7941	**0.8125**	0.7718	0.8177	**0.8218**	0.7420
AVERAGE	0.8186	0.7349	0.7527	0.8267	0.7416	0.7591	0.8038	0.7394	0.7517

TABLE IV. COMPARE WITH THE DREP AND DIVP ALGORITHMS

	L=100		
Datasets	**My**	**DREP**	**DivP**
Wine	**0.9722**	0.9658	0.9600
Aus	**0.8841**	0.8265	0.8374
Veh	**0.7824**	0.7537	0.7586
Iono	0.8310	0.8853	**0.9207**
Wf	**0.8630**	0.7902	0.7940
Sonar	**0.8124**	0.7929	0.8000
Rwine	**0.6125**	0.5592	0.5788

Gc	**0.7800**	0.7168	0.7434
Sb	**0.9349**	0.9260	0.9247
Eco	0.7941	**0.8026**	0.7692
AVERAGE	0.8267	0.8019	0.8087

TABLE V. COMPARE WITH PAIRWISE DIVERSITY MEASURES

	L=100					
Datasets	**My**	**P_1**	**P_2**	**P_3**	**P_4**	**P_5**
Wine	**0.9722**	0.9418	0.9321	0.9601	0.9378	0.9624
Aus	**0.8841**	0.8495	0.8768	0.8623	0.8523	0.8550
Veh	**0.7824**	0.7470	0.7764	0.7685	0.7347	0.7813
Iono	0.8310	0.8388	0.8450	**0.8732**	0.8458	0.8591
Wf	0.8630	0.8549	0.8480	0.8440	0.8600	**0.8750**
Sonar	**0.8124**	0.6190	0.7667	0.6667	0.6667	0.6192
Rwine	**0.6125**	0.5500	0.5906	0.5738	0.5840	0.5563
Gc	**0.7800**	0.7300	0.7450	0.7450	0.7200	0.7424
Sb	**0.9349**	0.9305	0.9261	0.9186	0.9239	0.9274
Eco	0.7941	0.7794	0.7794	**0.7958**	0.7352	0.7647
AVERAGE	0.8267	0.7841	0.8086	0.8008	0.7860	0.7943

TABLE VI. COMPARE WITH NON-PAIRWISE DIVERSITY MEASURES

	L=100					
Datasets	**My**	**M_1**	**M_2**	**M_3**	**M_4**	**M_5**
Wine	**0.9722**	0.9524	0.9638	0.9222	0.9166	0.9475
Aus	**0.8841**	0.8827	0.8695	0.8578	0.8650	0.8688
Veh	**0.7824**	0.7812	0.7705	0.7698	0.7741	0.7705
Iono	0.8310	**0.8732**	0.8309	0.7823	0.8391	0.8450
Wf	0.8630	0.8740	0.8500	**0.8970**	0.8437	0.8700
Sonar	**0.8124**	0.7380	0.6904	0.7380	0.7819	0.6904
Rwine	**0.6125**	0.5687	0.5437	0.5812	0.5718	0.5423
Gc	**0.7800**	0.7400	0.7342	0.7500	0.7650	0.7750
Sb	**0.9349**	0.9126	0.9346	0.9050	0.9313	0.9274
Eco	**0.7941**	0.7794	0.7352	0.6911	0.7647	0.7352
AVERAGE	0.8267	0.8102	0.7923	0.7894	0.8053	0.7972